

JAN 9 2024



EOM 232.01

Identify Types of Aircraft Engines

By:





Important!



You'll be able to identify aircraft engines



You'll learn more about their uses in different aircrafts



Helpful for aircraft maintenance or designing



Teaching Points



1

Explain That a
Powered Aircraft
Needs a Means of
Propulsion

2

Explain Combustion
in Rocket, Gas
Turbine and Piston-for
powered Engines

3

Explain the
Oxidization Process
for Different Types of
Engines

4

Identify Aircraft
and Associated
Engine Types



Teaching Point 1

Explain That a Powered Aircraft Needs a Means of Propulsion



A powered aircraft needs a **propulsion** to overcome drag and allow the wings to generate sufficient lift to overcome weight.



The propeller and jet engine are very closely related, providing thrust by accelerating air.

- The propeller accelerates a **large amount of air by a small amount**
- The jet engine accelerates a **smaller mass of air by a large amount**

The most common engine types used for aviation propulsion employ internal combustion and they include:

- rocket engines
- gas turbine jet engines
- reciprocating piston-powered engines.

Rocket Engines



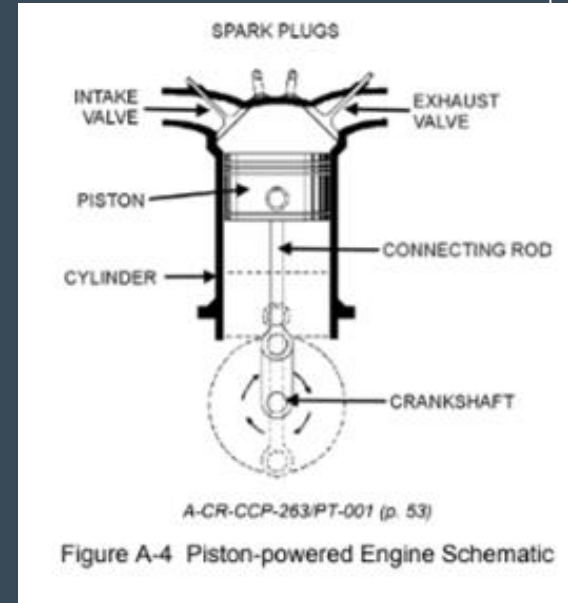
- Earliest engine used for Chinese fire arrows.
- Really effective method of propulsion
- Commonly used today with improvements for many applications including space exploration.
- Contained with their own oxidizer
- Advantageous in being able to function in a vacuum such as space.





Piston-powered engines

- Developed in the late nineteenth century
- Most common vehicle engine of all.
- Most complicated system of converting the chemical energy of fuel into motion
- Found in many places
 - Aircraft
 - Automobiles
 - Boats
 - Lawnmowers



Gas turbine engines

Any turbine converts the energy of moving liquid or gases into rotary motion to turn a shaft, such as a windmill.

Gas turbine jet engines have a simple ramjet that needs high velocity airflow to run. With the turbine, high pressure air can be generated so the engine can be started from a resting position.

They must be started first with a motor before being able to produce thrust because high pressure air is required for the engine to run

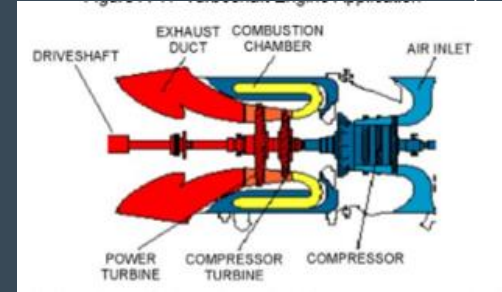
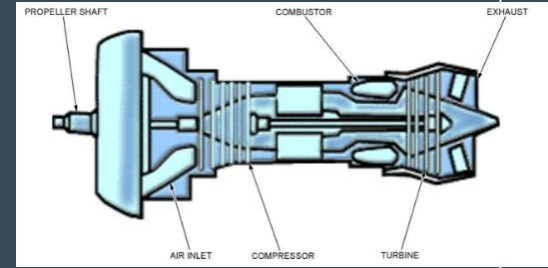
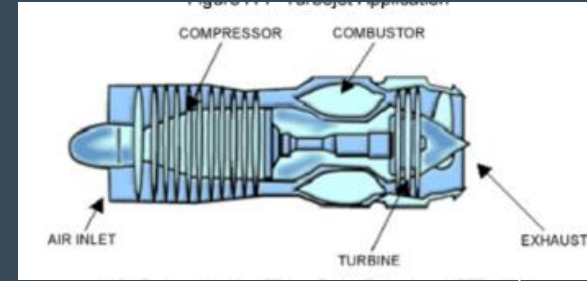


Gas Turbine variations (1/2)

A gas turbine jet engine that provides thrust, with no rotating shaft output, is a **TURBOJET** engine.

A gas turbine jet engine that provides thrust and also drives a propeller is a **TURBOPROP** engine.

A gas turbine engine that drives a helicopter rotor is usually a **TURBOSHAFT** engine. In a turboshaft helicopter engine, the output driveshaft is separate from the compressor turbine shaft so that engine speed is not tied to the helicopter's main rotor speed.



Gas Turbine Variations (2/2)

The most common variation of the gas turbine jet engine is the **TURBOFAN**, which is a hybrid of a turbojet and a turboprop.

The turbofan has a fan instead of a propeller that provides thrust. This allows the aircraft to go faster than normal propellers, while also reducing engine noise and being fuel efficient.

These properties make them very effective for commercial aviation.



TP1 Confirmation



QUESTIONS

Q1. Which engine type was the first to be used for propulsion?

Q2. What are the 4 variations of gas turbines mentioned here?

Q3. Why does a gas turbine jet engine need to have a starting motor?

because high-pressure air is needed to operate the engine.

Teaching Point 2

Explain Combustion in
Rocket, Gas Turbine and
Piston-powered Engines





All rocket, jet and piston-powered engines are internal-combustion engines because they all use burning fuel to generate power from expanding gases in a closed container.

When fuel is **oxidized** (reacted with oxygen) it gives off heat. The heat causes expansion of the gases that result from the oxidization.

- Very slow oxidization is simply **oxidization**
- Fast oxidization, it is referred to as **burning**.
- Very rapid oxidization is an **explosion**

All these processes result from fuel chemically combining with oxygen. The distinguishing characteristic between them is the speed of the reaction.



Usefulness of Combustion

By capturing the expanding hot gases of combustion in a tightly closed container, such as a piston-powered engine's combustion chamber or a gas-turbine jet engine combustor, the energy of the hot gases can be put to useful work.

A common method of using this power is the turning of a shaft, such as a turbine shaft or a crankshaft. That shaft can then be used to turn an aircraft's propeller or an automobile's wheels.

TP2 Confirmation



QUESTIONS

Q1. What causes burning gases to expand?

Q2. What is the difference between fuel burning and fuel exploding?

Q3. What is a common method of using power generated by combustion?

ANSWERS

Teaching Point 3

Explain the Oxidization
Process for Different
Types of Engines





Rocket Engine

The rocket has the simplest method of oxidization and carries its own fuel and oxygen and combines them in a closed container to generate the energy.

Fuel and the oxygen must be carefully calculated and loaded before launch since a rocket can only use the loaded fuel and oxygen.

The hot expanding gases that result from an explosion in the rocket's combustor are blasted out the back of the rocket at high speed to propel it upward.

Gas Turbine Engine (1/2)

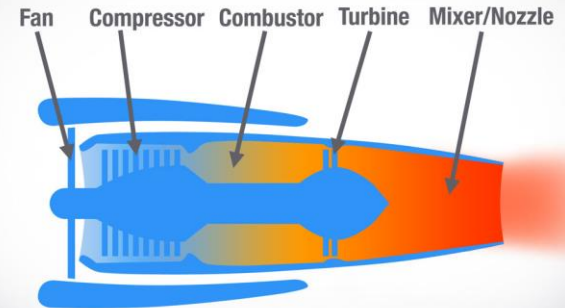
A gas turbine jet engine typically resembles a hollow cylinder with air sucked in the front and blasted out the back.

These can be seen under the wing of most airliners.

The fuel is stored in tanks, often in the wing.
The air, which contains oxygen, is sucked by a compressor fan



Parts Of A Turbofan



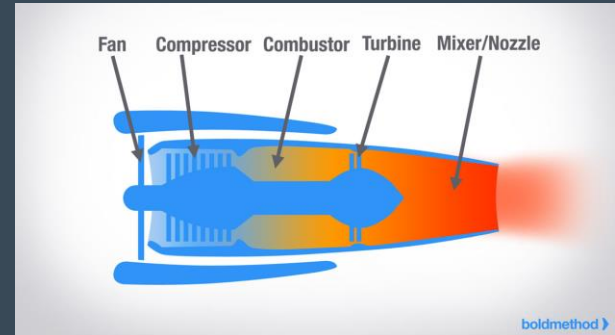
Gas Turbine Engine (2/2)



The fuel is combined with the pressurized air and the mixture is detonated in a container within the engine called a combustor.

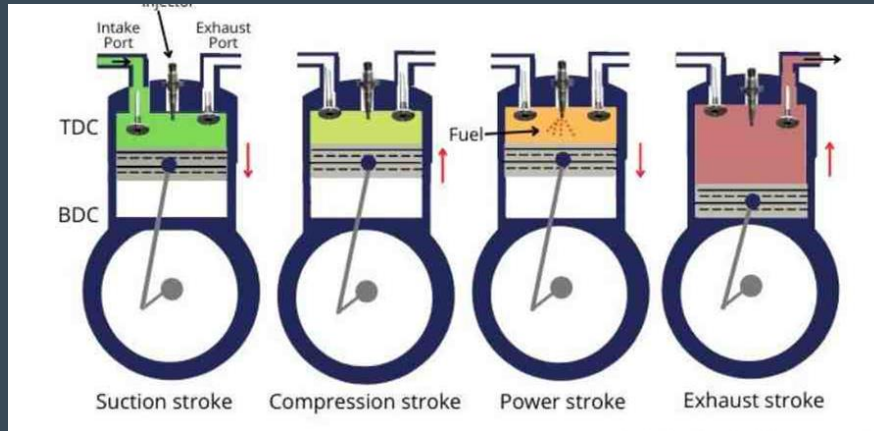
The hot expanding gases are blasted out the back of the engine to propel the aircraft forward.

A turbine sits beside the combustor and uses a portion of the energy produced to spin a shaft that drives the compressor fan to suck the air into the engine.



Piston-Powered Engine

The reciprocating piston-powered engine is the most common and intricate method of oxidization. In a four-stroke piston-powered engine, air is carefully mixed with small fuel droplets and then either sucked or injected into cylinders where the mixture is detonated to drive pistons in a cycle of intake, compression, power, and exhaust.



TP3 Confirmation



QUESTIONS

Q1. Where does a rocket get oxygen to burn fuel in outer space?

Q2. What is used to spin the turbine in a gas-turbine engine?

Q3. Where does a jet engine, such as a gas-turbine, get oxygen to burn fuel?

ANSWERS



Teaching Point 4

Identify Aircraft
and Associated
Engine Types



C-130 Hercules: four Allison T-56-A-7/15 turboprop engines.

The type of engines on an aircraft determined what job it was meant for: Jet-engine aircraft are generally seen as “strategic”, while turboprop-powered aircraft are “tactical”. Tactical transports are usually designed to operate on rougher, shorter, more primitive airfields than the facilities required by strategic transports.



CC-150 Polaris: two General Electric CF6-80C2A2 turbofan engines.

This strategic lift aircraft's primary role is long-range transport of people and equipment, up to 194 passengers or 32 000 kg of cargo. They have participated in operations supporting the CF, NATO and numerous UN and Red Cross initiatives.





Harvard North American T-6J: one nine-cylinder Pratt & Whitney radial engine.

It was assigned to No. 1 Flying Instructors School which still exists today as Canadian Forces Flying Instructors School. The training aircraft was then upgraded from the Harvard to the CT-114 Tutor. The planes were flown by the Golden Hawks, an aerobatic display team of the 1950s.



CT-114 Tutor: one General Electric J85-CAN-40 turbo jet engine.

CT-114 Tutors were flown by the Snowbirds, Canada's world famous aerial acrobatic team. The Tutor, a Canadian designed and produced single-engine subsonic jet trainer that entered service in the mid-1960s, was used for basic and advanced pilot training





CH-146 Griffon: one Pratt & Whitney PT6T-3D turboshaft engine.

As Canada's Utility Transport Tactical Helicopter (UTTH), the Griffon provides a reliable and cost-effective capability to conduct several operations such as airlift of equipment and personnel, surveillance and reconnaissance, casualty evacuation, search and rescue, and more.



End of Lesson Confirmation

QUESTIONS

Q1. Why are the rocket, gas-turbine and piston-powered engines all internal combustion engines?

Q2. What is used to spin the turbine in a gas-turbine engine?

Q3. What causes burning gases to expand?

ANSWERS



To recap we looked at:

- Types of engines:
 - Rocket
 - Gas turbines
 - Piston-powered engine
- Combustion in aircrafts
- Oxidization processes in each engine
- Engine applications in different aircrafts

Thanks for listening!

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